ALIF FAJRI ANTARIKSA

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PROFILE

Mechanical Engineering master's student at Institut Teknologi Bandung (ITB) with a background in Manufacturing Design Engineering and professional experience in mechanical design and product development at PT Polylamina Teknologi Indonesia and PT Tri Graha Sealisindo. Also participated in the Taiwan Experience Education Program (TEEP) at National Chin-Yi University of Technology, Taiwan. Experienced in mechanical system design, manufacturing engineering, technical documentation, and structural analysis using Computer-Aided Design (CAD) and Finite Element Analysis (FEA). Currently developing expertise in additive manufacturing, material characterization, structural optimization, and biomechanical engineering, with a particular interest in research and development (R&D).

SKILLS

Technical Skills

- Mechanical and Machine design
- 3D and 2D design using CAD software
- Structural Analysis and Optimization using FEA
- Additive Manufacturing (FDM/FFF)
- Materials and Manufacturing Processes
- Engineering documentation according to international standards
- Product Development

Soft Skills

- Leadership
- Communication Skills
- Critical thinking
- Adaptability

CAD Software

- Solidworks
- Autodesk Inventor
- PTC Creo
- Siemens NX
- FreeCAD
- AutoCAD
- ZWCAD
- Open CAD Studio

Simulation Software

- Ansys Simulation
- Abaqus
- Solidworks Simulation
- FreeCAD FEM

WORK AND INTERNSHIP EXPERIENCE

PT Polylamina Teknologi Indonesia **December 2023 – December 2024**

Design Engineer

- Performed structural strength simulations for interior components of the Bandung Al-Mumtadz Mosque project.
- Developed the façade gutter system for the Bank Indonesia – IKN building project, from preliminary design through engineering drawings.
- Designed FRP mold tooling and prepared its engineering drawings for the Bank Indonesia – IKN façade project.

- Conducted structural strength analysis of FRP façade brackets to evaluate their mechanical performance.
- Prepared engineering drawings, including raw material drawings, tooling drawings, module drawings, FRP façade partition and shop drawings for FRP components.
- Prepared engineering data to support project cost estimation and quotation processes.

Taiwan Experience Education Program **September 2022 – January 2023**
Internasional Internship

- Received training on semi-conductors, especially on the manufacturing process.
- Conducted laboratory research and studies.
- Visited factories & manufacturing technology centres in Taiwan.
- Introduction to Taiwanese culture.

PT Tri Graha Sealisindo
Design Engineer – Internship **March 2022 – July 2022**

- Designed a 4" – 18" Ring Spiral Wound Gasket Vending machine.
- Designed a plate shearing knife sharpening machine.
- Designed an anchor bending tool.
- Modified the design of the SWG outer ring press machine.
- Prepared technical drawings of maintenance components.
- Prepared a cost estimate of the machine design.
- Repaired the saddle of the circular Expansion Joint Fabric machine.

CERTIFICATION

Engineer Desain Mesin / Mechanical Design Engineer **October 2023 – October 2026**
National Professional Certification Agency – POLMAN Bandung Professional Certification Body.
Registration Number: LOG 1363 00141 2023

Basic Programming PLC & HMI Mitsubishi **May 2022**
Engineering Academy – Anak Teknik Indonesia
Registration Number: PLM-VI-22-ATI-216735080

Project Management & Scheduling With Microsoft Project **October 2020**
Registration Number – Anak Teknik Indonesia

EDUCATION

Institut Teknologi Bandung **2025 – Present**
S2 – Master of Mechanical Engineering

- Ongoing Research: Design and optimization of 3D-printed transtibial prosthetic sockets using Taguchi and Response Surface Methodology (RSM) with lattice structure.
- Research Focus: Additive Manufacturing, Structural Optimization, and Biomechanical Engineering.

Bandung Polytechnics for Manufacturing **2019 – 2024**
D4 – Manufacturing Design Engineering Technology GPA: 3.16/4.00

- Final Project "Design of Elephant Grass and Corn Stalk Cutting Machine with 2-3 cm Cut as Silage Material for Cattle Feed".
- Participated in the Taiwan Experience Education Program (TEEP) international internship programme at National Chin-Yi University of Technology.
- Participated in the Industrial Practice Programme at PT Tri Graha Sealisindo.
- Participated in campus organisations at the department level, namely the Design Engineering Student Association.

ORGANIZATION

Design Engineering Student Association **2021 – 2022**
Legislative Chairman

- Take the lead in legislative meetings to discuss policies, laws, and important legislative issues for the organisation.
- Be the main leader in the implementation of supervision of the executive body of the organisation on the review of laws that have been mutually agreed upon.
- Coordinating and managing the organisation's legislative members.
- Responsible for all legislative activities in the organisation.
- Decision/policy maker in the legislative field of the organisation.

Design Engineering Student Association **2020 – 2021**
Vice Chair of the Organization Knowledge Development Division

- Support the division chairperson in planning, organising, and supervising the division's work programme.
- Representing the chairperson in all divisional work programmes when absent.
- Being a division representative at internal organisational meetings together with the division chairperson.
- Become a bridge between the chairman and division members related to the aspirations of division members.

LANGUAGE

Bahasa Indonesia – Native Speaker
English – Intermediate